WEEK 9 / HOMEWORK REPORT

Web Performance Optimization

Team Project: Memory Match | GitHub Pages | Lighthouse Performance audit

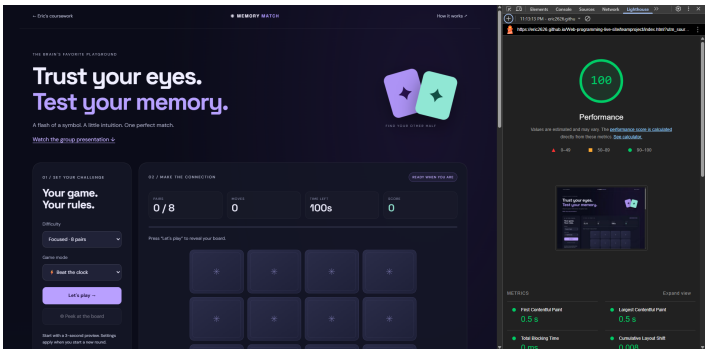
BEFORE / BASELINE

100 / 100

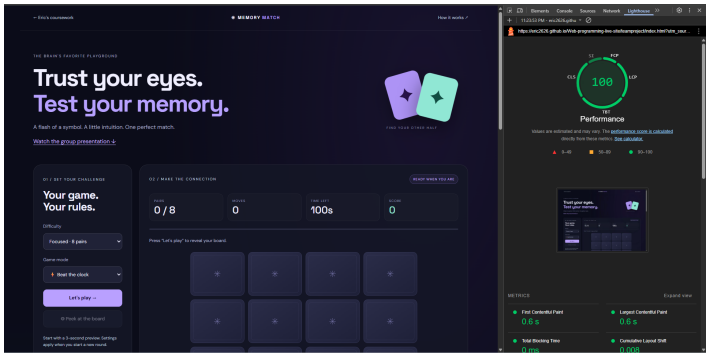
AFTER / OPTIMIZED

100 / 100

Lighthouse screenshots



Before changes



After five code improvements

Measured results (from the screenshots)

Metric	Before	After
Performance score	100 / 100	100 / 100
First Contentful Paint	0.5 s	0.6 s
Largest Contentful Paint	0.5 s	0.6 s
Total Blocking Time	0 ms	0 ms
Cumulative Layout Shift	0.008	0.008

Result

The project already scored 100 before optimization and remained at 100 afterward.

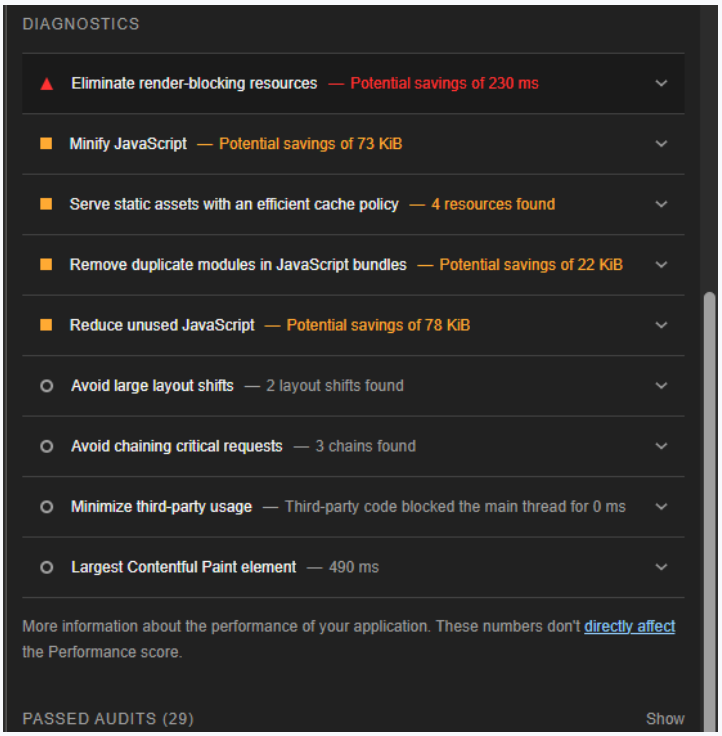
A higher score was not possible; timing differences between audits are normal.

Five targeted code improvements were still applied and documented.

BASELINE REVIEW

Original Lighthouse Diagnostics

Screenshot taken before the new team-project changes



Original findings

01 **Render-blocking resources**
Potential 230 ms saving

02 **Minify JavaScript**
Potential 73 KiB saving

03 **Inefficient cache policy**
4 resources

04 **Duplicate JavaScript modules**
Potential 22 KiB saving

05 **Unused JavaScript**
Potential 78 KiB saving

06 **Layout shifts**
2 shifts reported

How the baseline informed the changes

The report showed network-loading and render-path opportunities even though the overall Performance score was already perfect.

We reduced font-request dependency, gave the browser an earlier module hint, stopped the presentation video from loading metadata up front, and allowed off-screen presentation sections to skip initial rendering work.

Note: The original diagnostics do not prove that all six issues were fixed.

IMPLEMENTATION

Five Optimizations Applied

Code snippets are from the updated teamproject/index.html and teamproject/style.css

1 | Preconnect to font services

```
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
```

Starts connections for Google Fonts earlier.

2 | Load fonts directly from HTML

```
<link rel="stylesheet" href="https://fonts.googleapis.com/css2?...">
/* Removed the previous @import from style.css */
```

Removes an extra CSS import step.

3 | Preload the game engine module

```
<link rel="modulepreload" href="engine.mjs?v=2">
```

Lets the browser discover the game dependency sooner.

4 | Avoid loading video metadata at startup

```
<video controls playsinline preload="none" ...>
```

Delays an unnecessary media request until needed.

5 | Defer off-screen section rendering

```
.team-info,#recording{content-visibility:auto;
  contain-intrinsic-size:auto 440px}
```

Skips initial painting for off-screen presentation content.

Reflection

The video preload change was the most useful practical improvement because the recorded presentation is not needed when the game first opens. The score stayed at 100, so the audit does not establish a measurable score gain from this change.